When does compression cost the task and what keeps the VAE alive?

β-VAE on TCGA downstream AUC on z · cancer type and sex

STRUCTURE

The setting, the latent space and the metrics, then three tasks with the same pattern

Setting

- Question, data, model and what the rate contains
- Part 1 · The latent space under compression (2D views)
- Part 2 · The metrics: 56

Three tasks, then two consequences

- A · Downstream: which metrics explain the AUC?
- B · Collapse: which metrics explain the rate?
- C · Both: what separates useful metrics from compression?

QUESTION AND SETUP

Compress as much as possible without losing what the task needs

β weights the KL penalty - the achieved rate R bounds MI(X;Z) and shows posterior collapse

As R becomes small, the conditional posterior approaches the prior.

Which measurements of the latent space tell us where we are, and more than the rate itself?

Component	Setup
Data	TCGA RNA + methylation, 2000 features, 707 test samples
Model	AUTOENCODIX Varix, d = 2, 4, 6, 8, 12, 16
Sweep	$\beta/d = 0, 5{\cdot}10^{-6} \ldots 0.015 \cdot 4$ seeds · 90 epochs
β/d	KL is averaged over d, so β/d is the weight per nat of rate
Measurements	56 metrics; μ, z or posterior parameters; kNN k = 20
Downstream	AUC on z (one draw), logistic regression
Tasks	cancer type (8 classes) · sex in mixed cancer types

TRAINING PARAMETERS

Training settings: β, d and training seed vary

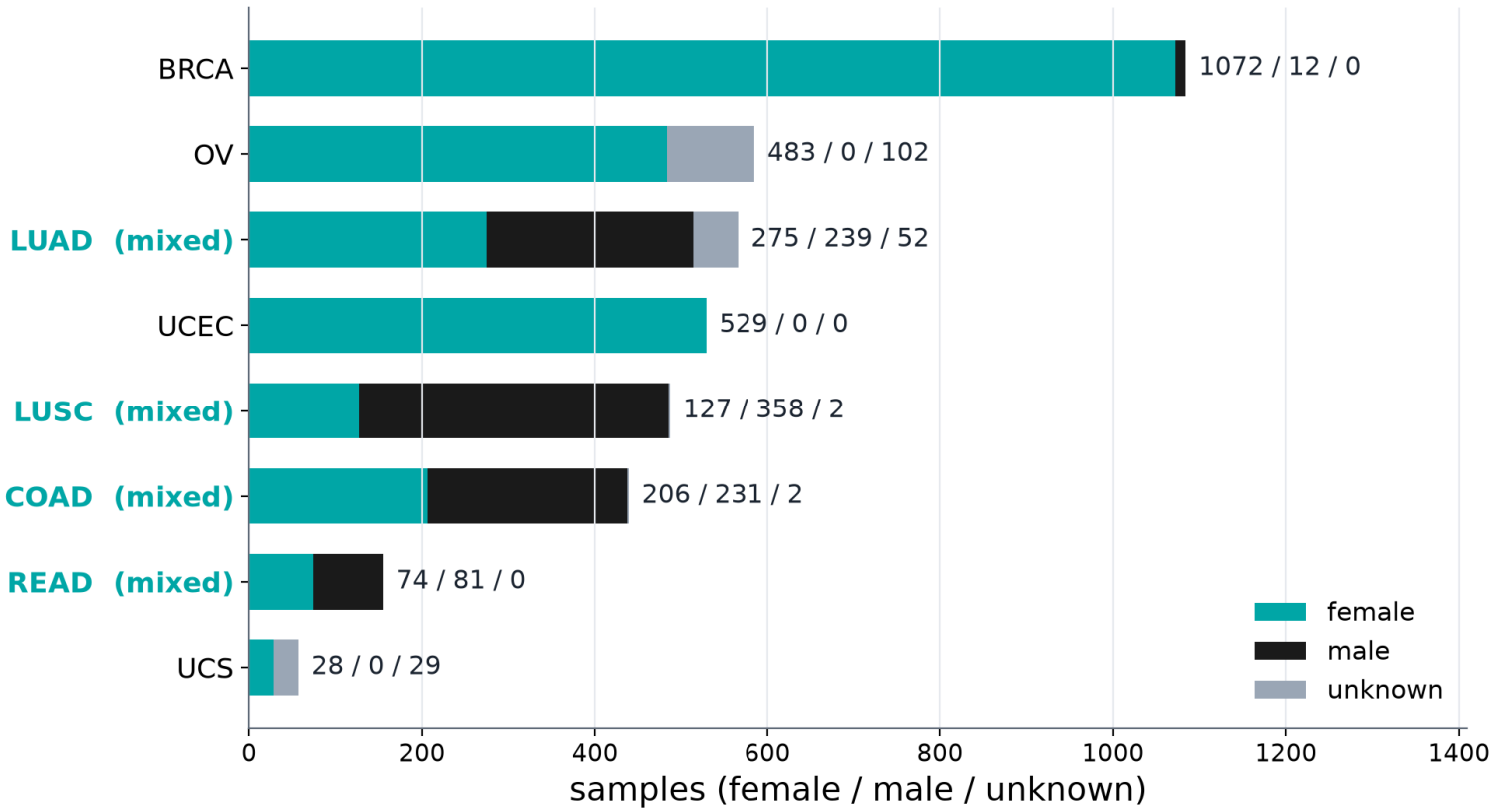
	Value
Dataset	TCGA, split seed 2
Latent dimension d	2, 4, 6, 8, 12, 16
β/d	0, $5{\cdot}10^{-6}$, $1.25{\cdot}10^{-5}$, $2.5{\cdot}10^{-5}$, $5{\cdot}10^{-5}$, 10^{-4}, $2.5{\cdot}10^{-4}$, 10^{-3}, $4{\cdot}10^{-3}$, 0.015
Seeds	13, 17, 23, 31
Learning rate	$5{\cdot}10^{-4}$
Epochs	90
Regulariser	KL, mean reduction, no annealing

	Value
Model	Varix, RNA + methylation
Preprocessing	min-max, 2000 features
Split	70 / 10 / 20 %
Architecture	3 layers, factor 4, dropout 0.1
Optimiser	AdamW, weight decay 0.01
Batch size	32
Reconstruction	MSE, float32

6 dimensions × 10 β/d values × 4 seeds = 240 runs

DATA · SEX PER CANCER TYPE

Only four cancer types contain both sexes; the sex task is evaluated only there



682 / 909

women / men in COAD, LUAD, LUSC and READ: the samples of the sex task

56

samples with unknown sex in these four types (left out)

BRCA, OV, UCEC and UCS contain almost only women: there, sex would just repeat the cancer type.

All samples: 2794 women, 921 men, 187 unknown.

How computed: counts over all splits from the clinical table; no model involved.

WHAT THE RATE CONTAINS

Rate bounds information; it does not identify the retained task signal

$$R = E_x\, KL(q(z|x) \,\|\, p(z)) = I(x;z) + TC(z) + \Sigma_j\, KL(q(z_j) \,\|\, p(z_j))$$

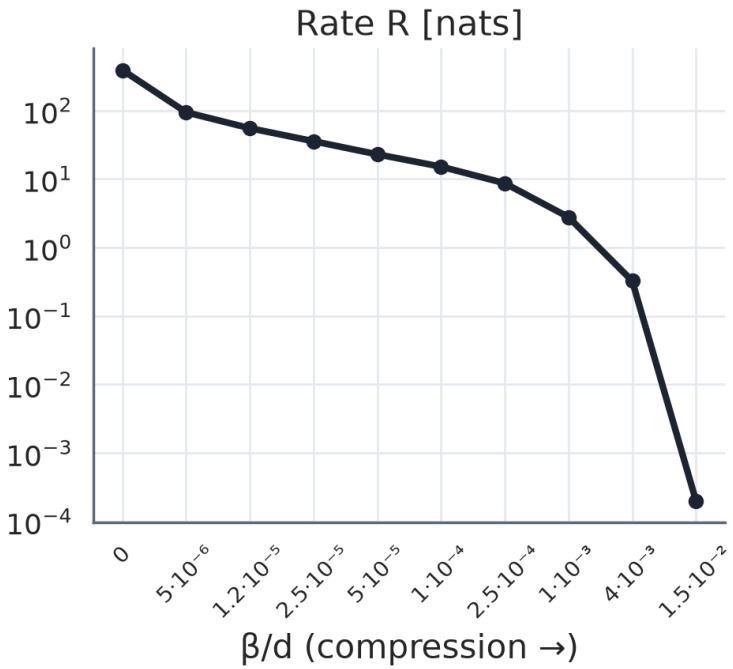
Structure and task utility therefore require separate measurements.

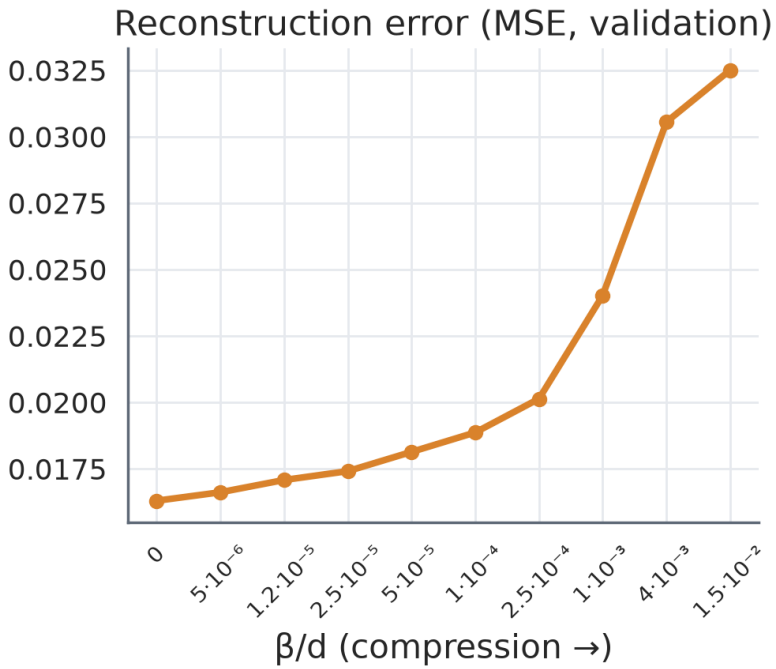
- $I(x;z)$: information about x carried by z. The other two terms are ≥ 0, so $R \geq I(x,z)$: the rate is an upper bound on information.
- $TC(z) = KL(q(z) \,\|\, \prod q(z_j))$: dependence between the latent axes (total correlation).
- $\Sigma\, KL(q(z_j) \,\|\, p(z_j))$: how far each latent marginal is from the prior N(0, 1).
- Equal rate can retain different neighbourhoods, classes or directions.

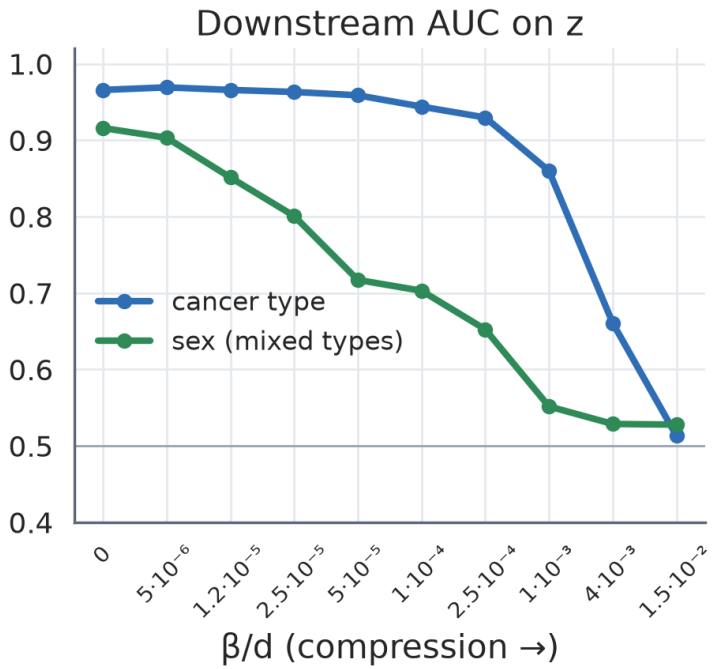
MAIN RESULT

Across β · AUC on z (one draw)

As β grows, the rate falls and reconstruction and the downstream tasks degrade at different points







388 → 2.8 nats

median rate, β/d = 0 → 10^{-3}

0.016 → 0.024

reconstruction MSE, same range

0.86 · 0.55

AUC at β/d = 10^{-3}: cancer type · sex

How computed: each β a separately trained model · median over 20 trajectories (d = 4, 8, 12, 16 × 4 seeds) · rate and reconstruction MSE on the validation split · AUC on z (one draw): cancer type validation (AUTOENCODIX), sex valid + test (COAD, LUAD, LUSC, READ).

PART 1 · THE LATENT SPACE

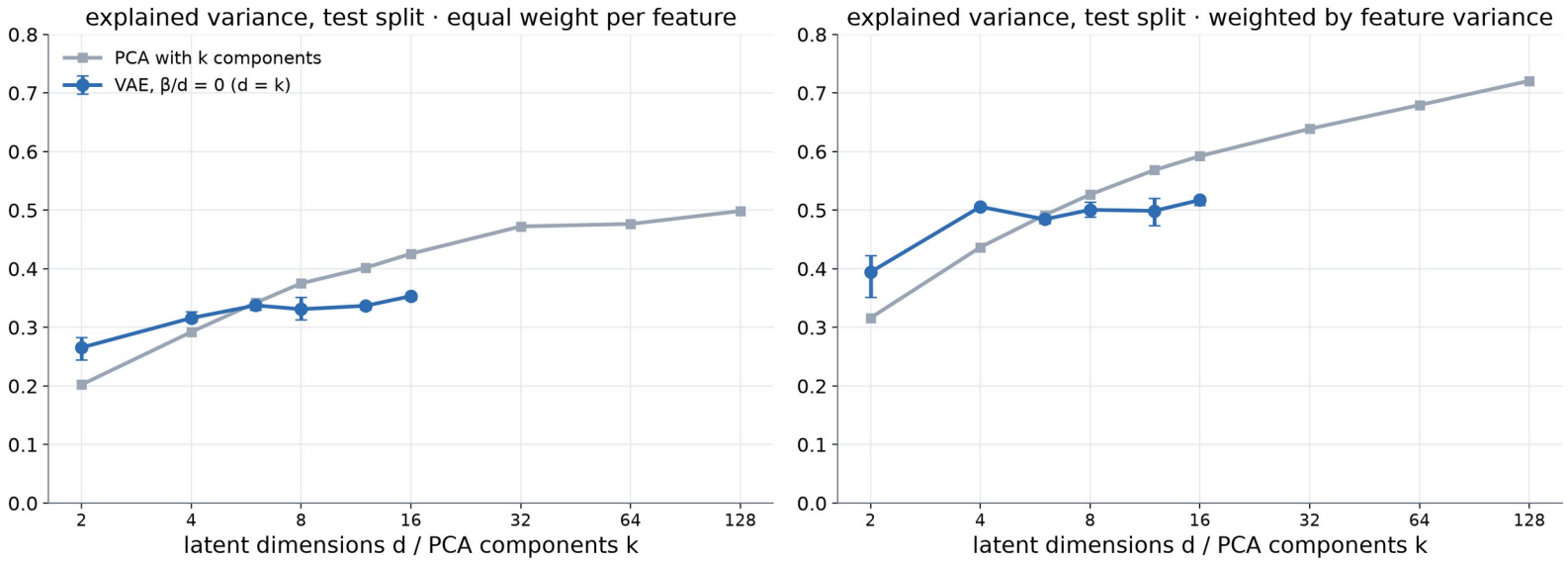
What does the latent space look like under compression?

Observed: cancer-type separation persists longer than the mixed-type sex signal.

RECONSTRUCTION

Same β (β/d = 0) · test split

The VAE reconstructs about a third of the variance, close to PCA with as many dimensions



0.33 vs 0.38

explained variance at d = 8: VAE vs PCA with 8 components (equal weight per feature)

0.50

even PCA with 128 components: much of the per-feature variance is hard to compress

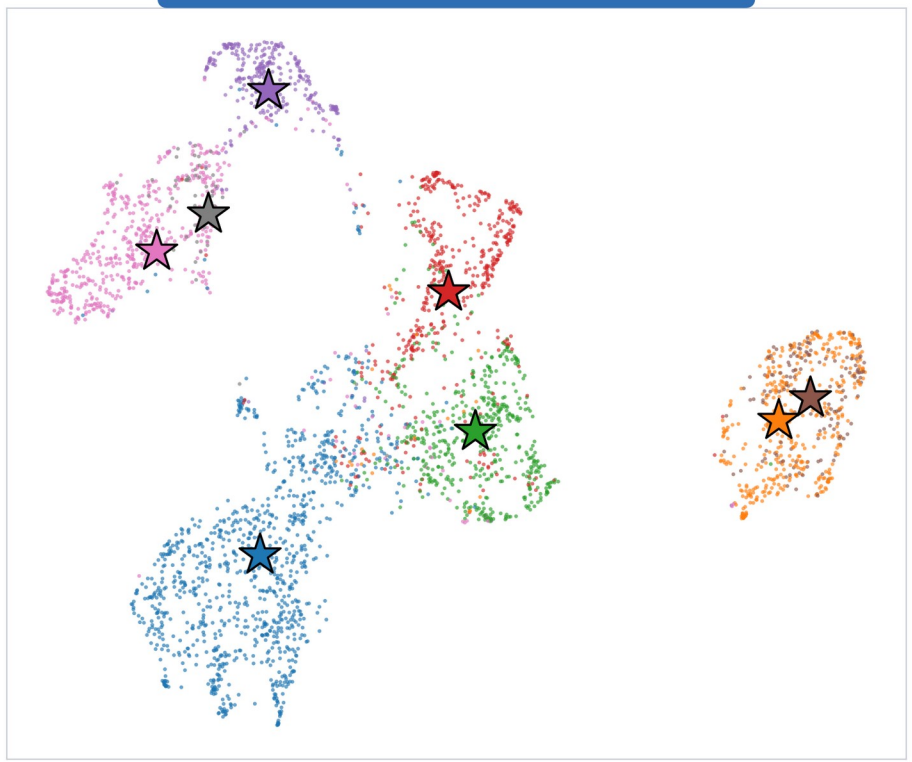
- Next: establish better convergence for next trainings and give model more opportunities to develop “good” latent space

How computed: VAE at β/d = 0, median over the 4 seeds per d, bars = IQR · PCA fit once on the training split, k = d components · both scored on the test split (explained variance, equal weight per feature).

One model, one β · z · UMAP

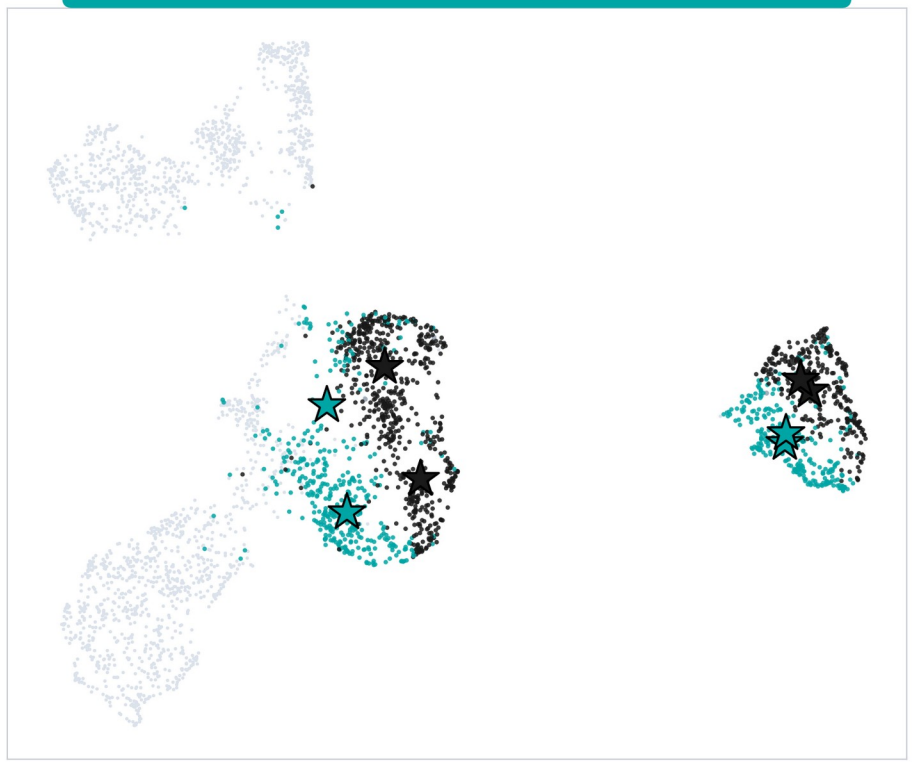
At the best AUC, both tasks are visible: cancer types as clusters, sex within the mixed types

TASK 1 · CANCER TYPE · AUC 0.975



BRCA COAD LUAD LUSC OV READ UCEC UCS

TASK 2 · SEX (mixed cancer types) · AUC 0.979

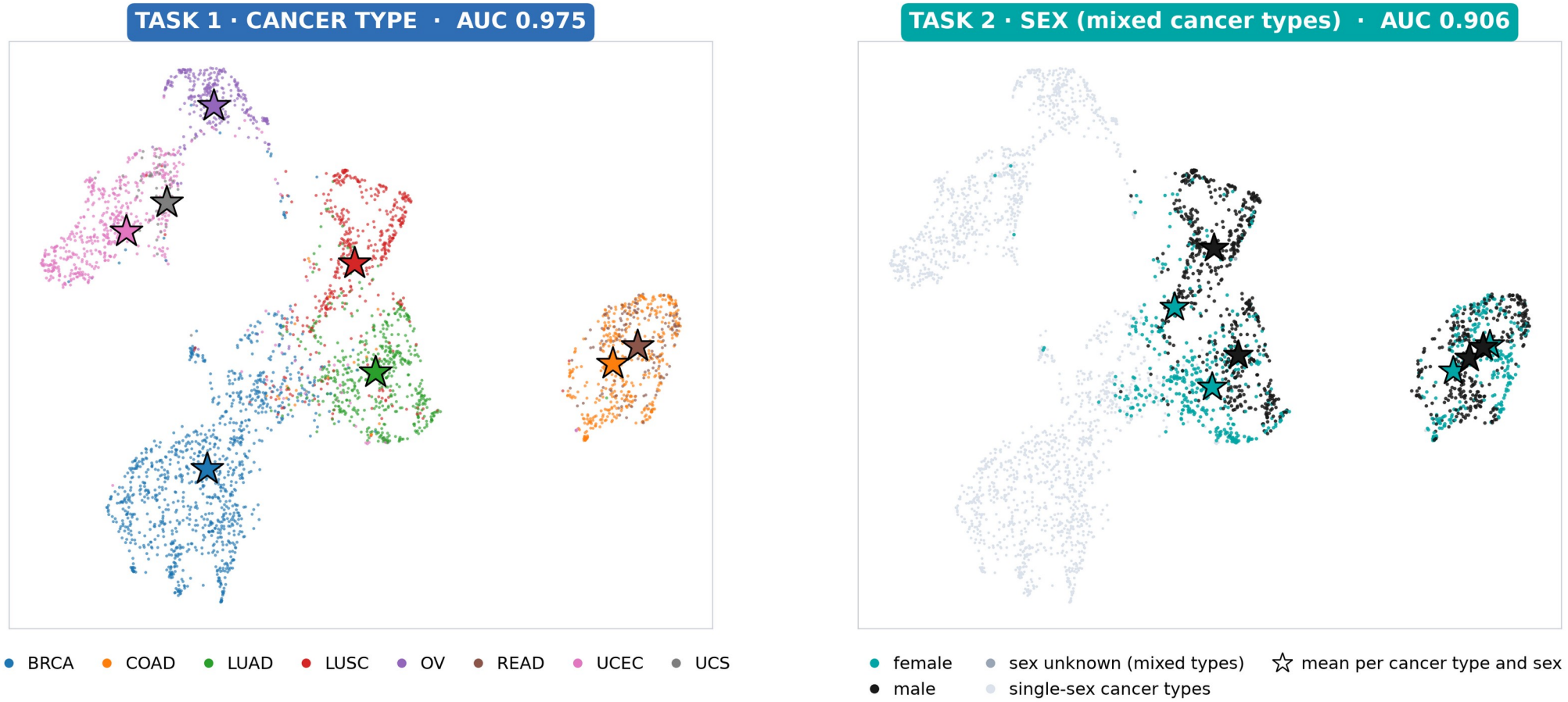


female male sex unknown (mixed types) single-sex cancer types ☆ mean per cancer type and sex

Cancer type: d = 8, seed 31, β/d = $1.25 \cdot 10^{-5}$ · AUC 0.975 Sex (mixed types): d = 16, seed 17, β/d = $5 \cdot 10^{-6}$ · AUC 0.979

One model, one β · z · UMAP

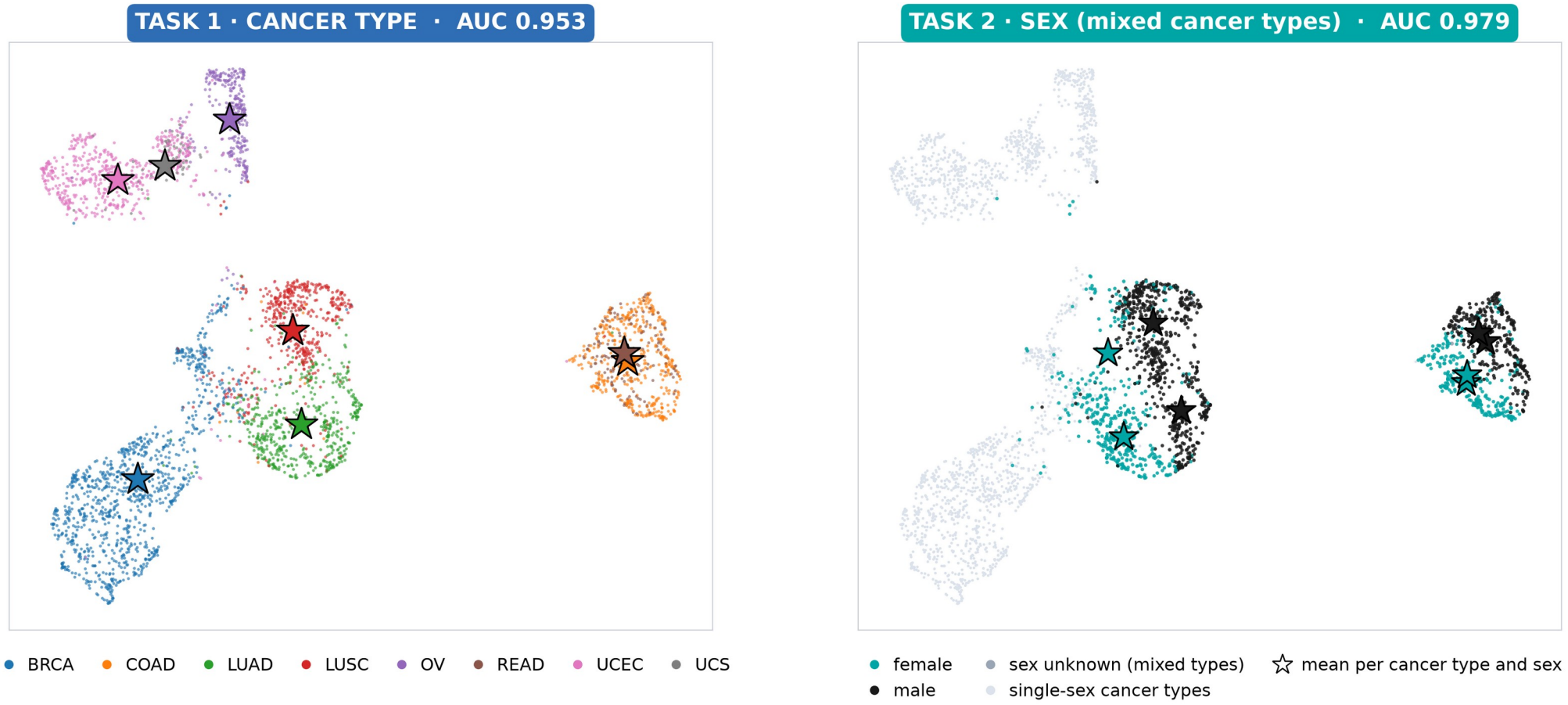
The model with the best cancer-type AUC: sex in the same latent space



One model for both panels: d = 8, seed 31, β/d = $1.25 \cdot 10^{-5}$ · cancer-type AUC 0.975 · sex AUC 0.906

One model, one β · z · UMAP

The model with the best sex AUC: cancer types in the same latent space

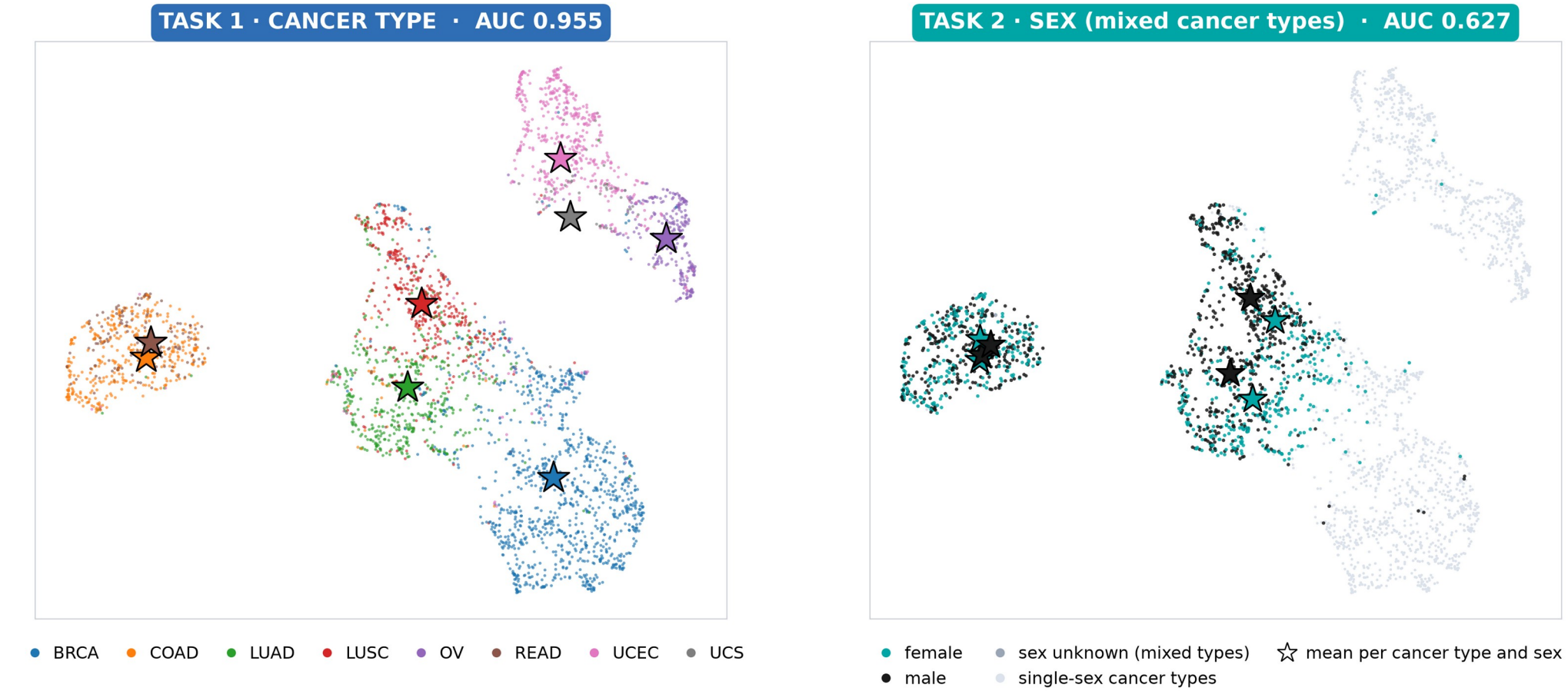


One model for both panels: d = 16, seed 17, β/d = $5 \cdot 10^{-6}$ · cancer-type AUC 0.953 · sex AUC 0.979

DOWNSTREAM TASKS IN 2D · CANCER TYPE KEPT, SEX LOST

One model, one β · z · UMAP

In the same model, cancer types are still separated while sex is already mixed

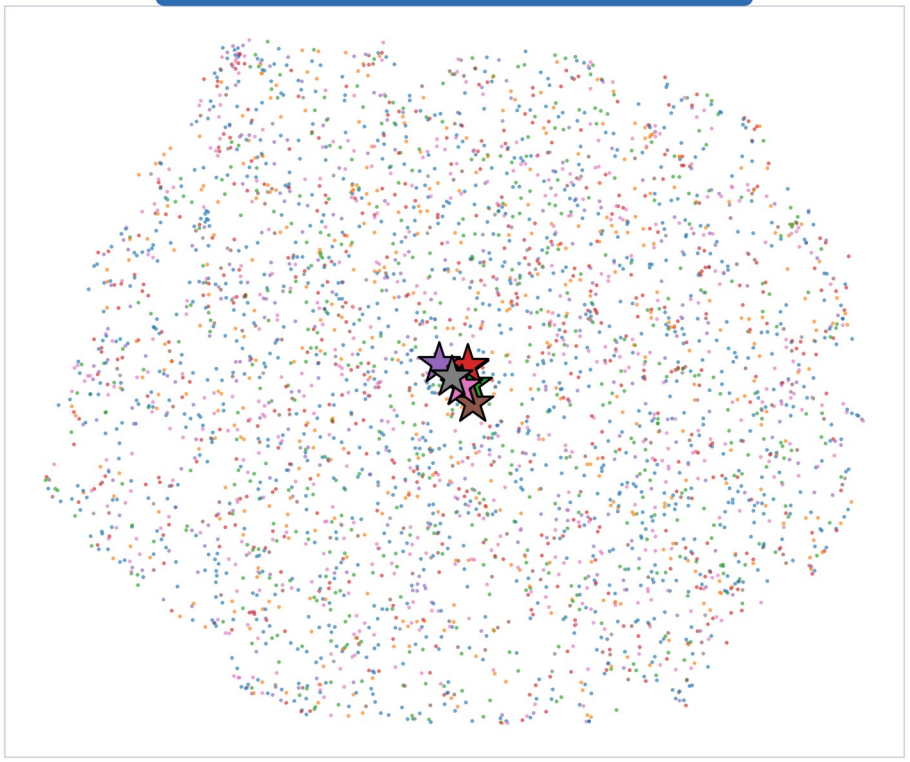


One model for both panels: d = 4, seed 17, β/d = $2.5 \cdot 10^{-5}$ · cancer-type AUC 0.955 · sex AUC 0.627

One model, strongest β · z · UMAP

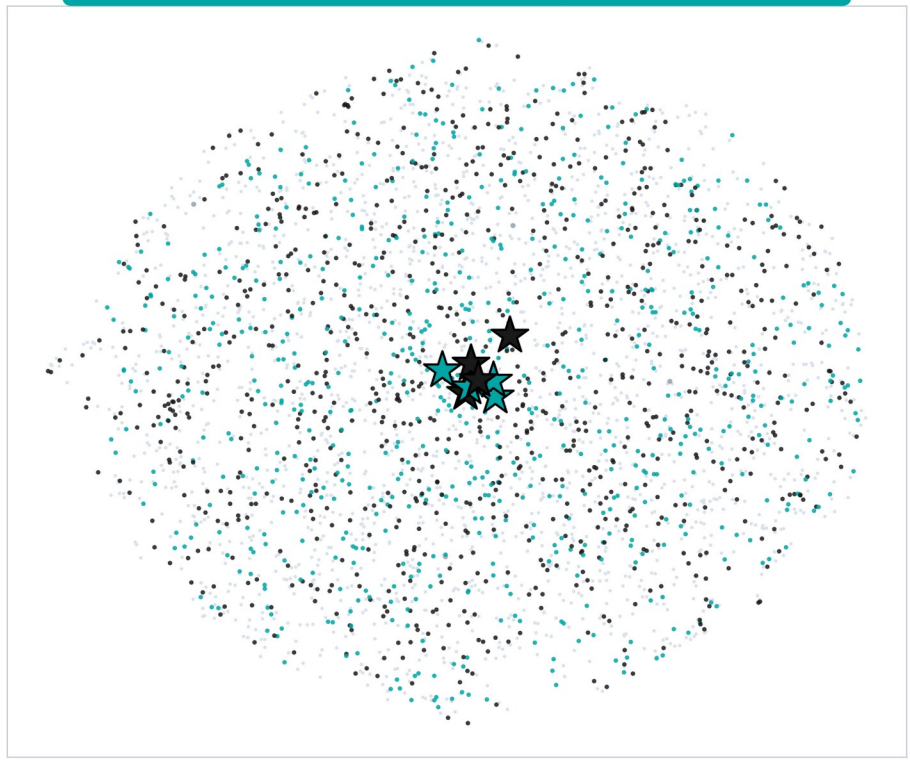
At posterior collapse, both task patterns are gone

TASK 1 · CANCER TYPE · AUC 0.501



BRCA COAD LUAD LUSC OV READ UCEC UCS

TASK 2 · SEX (mixed cancer types) · AUC 0.529



female male sex unknown (mixed types) single-sex cancer types ☆ mean per cancer type and sex

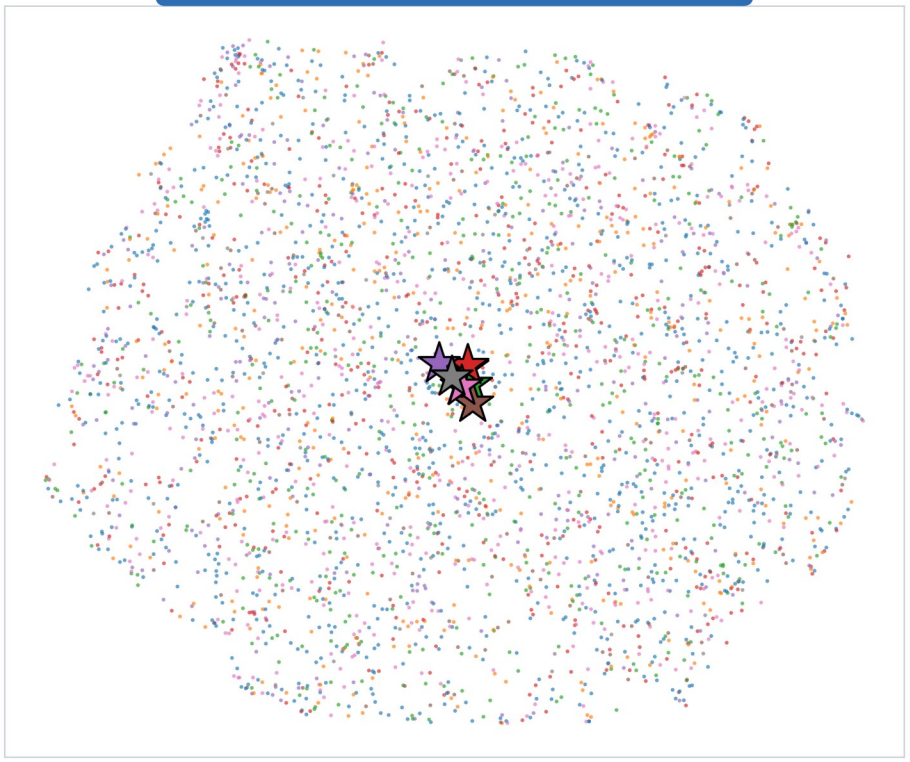
Cancer type: d = 8, seed 31, β/d = $1.5 \cdot 10^{-2}$ · AUC 0.501

Sex (mixed types): d = 16, seed 17, β/d = $1.5 \cdot 10^{-2}$ · AUC 0.529

One model, strongest β · z · UMAP

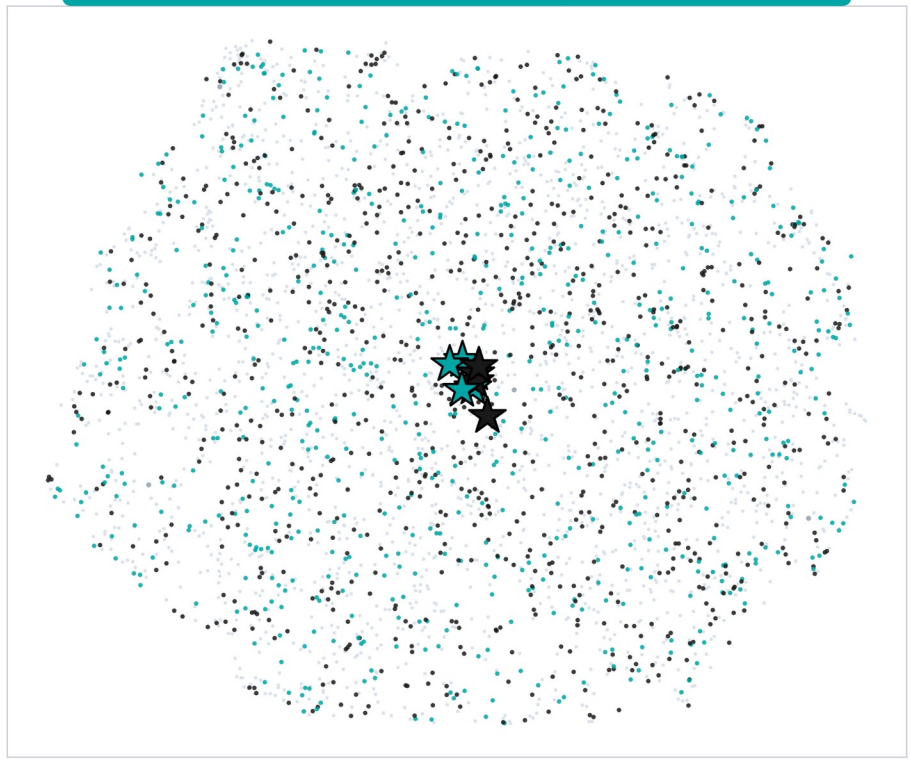
The best cancer-type model at collapse: both tasks gone in the same latent space

TASK 1 · CANCER TYPE · AUC 0.501



BRCA COAD LUAD LUSC OV READ UCEC UCS

TASK 2 · SEX (mixed cancer types) · AUC 0.520

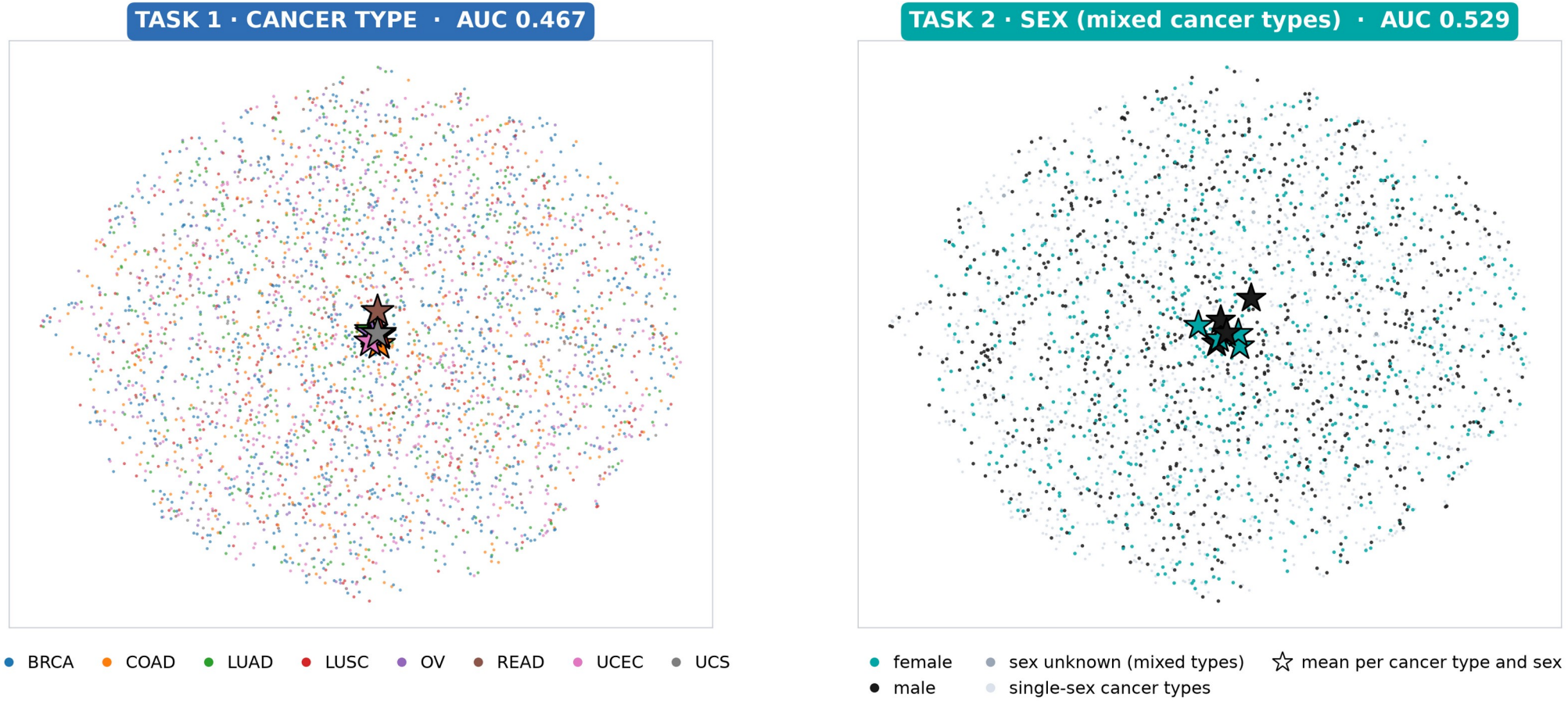


female male sex unknown (mixed types) single-sex cancer types ☆ mean per cancer type and sex

One model for both panels: d = 8, seed 31, β/d = $1.5 \cdot 10^{-2}$ · cancer-type AUC 0.501 · sex AUC 0.520

One model, strongest β · z · UMAP

The best sex model at collapse: both tasks gone in the same latent space



One model for both panels: d = 16, seed 17, β/d = $1.5 \cdot 10^{-2}$ · cancer-type AUC 0.467 · sex AUC 0.529

PART 2 · THE METRICS

What do we measure?

56 metrics describe μ, sampled z or preservation of X

THE METRICS · OVERVIEW

Mostly μ; z for intrinsic dimension and covariance TC; kNN k = 20

56 metrics/evaluation in seven families

Family · number	What it measures	Main examples
Geometry · 17	preservation, x vs. μ: are distances and neighbourhoods of x kept (ZADU)	steadiness, stress, trustworthiness, LCMC, continuity
Graph comparison · 4	preservation, x vs. μ: does the k-NN graph of μ match that of x	partition NMI / ARI, hub identity
TDA · 6	H0/H1 of μ, possible diagram comparisons with X after separate diameter scaling	H1 Wasserstein, H1 landscape, H0 persistence entropy
Graph · 22	the k-NN graph of μ alone: clustering, communities, hubs, paths	graph clustering, community size / entropy, hubness, modularity
Dimensionality · 4	how many directions μ or z really use	effective rank (+ fraction), intrinsic dim. of z (MLE, TwoNN): rate trackers
Curvature · 2	Forman–Ricci curvature of the latent graph	Forman mean / std
Information · 1	Gaussian covariance approximation to dependence in sampled z	total correlation (AUTOENCODIX): follows the rate

Geometry and graph comparison measure (X,μ) preservation. TDA includes comparisons and μ-only descriptors. Graph descriptors and effective rank describe μ; intrinsic dimension and covariance TC use z.

METRICS AND THEIR RATIONALE (1)

Latent space, k = 20

Preservation: what structure does the mean representation retain?

Metric	What it measures	Why it matters	Source
Trustworthiness · continuity	false neighbours in the latent space · missing neighbours	mixing of groups vs. tearing apart	
LCMC	overlap of k-nearest neighbourhoods between data and latent	local structure, k-NN based downstream	
Stress (scale-norm., non-metric)	distortion of all pairwise distances	global arrangement of groups	
Steadiness · cohesiveness	whether clusters stay clusters in both directions	cluster reliability	
Partition NMI / ARI	agreement of k-NN graphs and their communities	graph-level structure	
Persistent homology (Wasserstein, landscapes)	loops and components across scales	shape and topology	

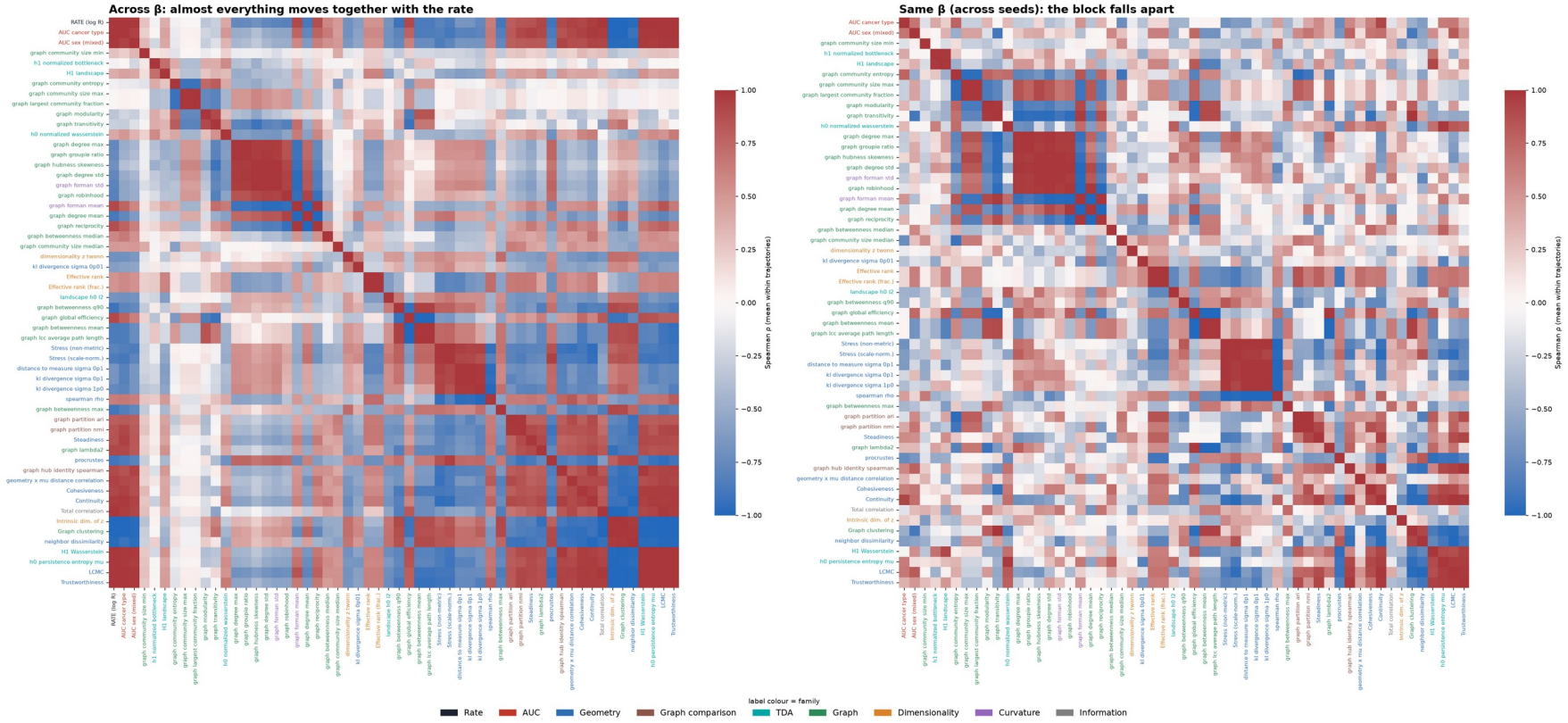
Latent space (z dim., TC: one z draw)

The latent space on its own: graph, dimension, information and collapse

Metric	What it measures	Why it matters	
Degree, hubness, clustering	shape of the k-NN graph of the latent space	hubs and crowding appear under compression	
Forman curvature	how edges spread or bottleneck the graph	basic graph-geometric signal	
Intrinsic dimension (MLE, TwoNN)	dimension of the data manifold in the latent space	how many directions are still used	
Effective rank	spread of the latent spectrum	dimensional collapse	
Total correlation	dependence between latent axes	disentanglement-type signal	
Rate · rate per dimension	information budget in the latent space	the compression itself	
Active units · ε-δ collapse	dimensions with variance or KL above a threshold	posterior collapse	

Left: across β · right: same β

Over all , the metrics move as one block with the rate - at equal betas the block falls apart



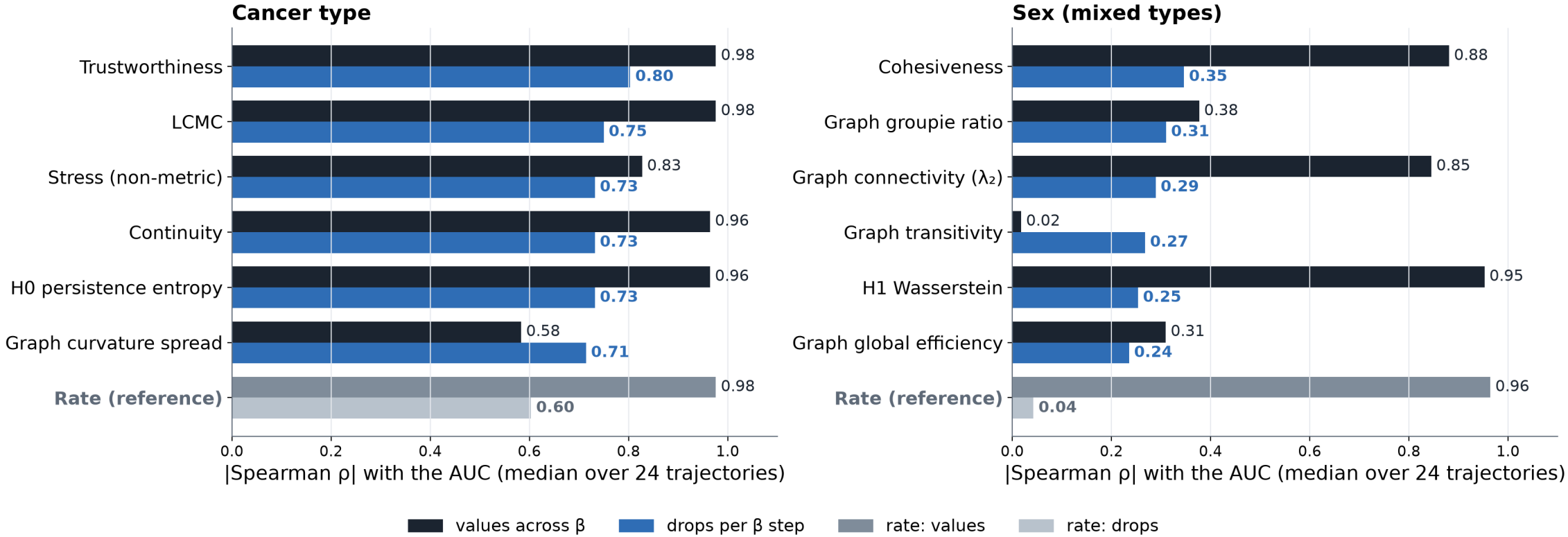
How computed: left, Spearman ρ between metrics within each trajectory (one d and seed, across β), averaged over all trajectories (Fisher z) · right, Spearman ρ across the 4 seeds within each cell of equal d and β, averaged over the 48 cells.

TASK A · DOWNSTREAM (AUC ON Z)

Which metrics explain the downstream task?

Across β · values and drops

Neighbourhood metrics have their largest drops at the same β steps as the cancer-type AUC; for sex no metric does



Dark: do metric and AUC fall together across β? Almost always, the rate too (0.98).

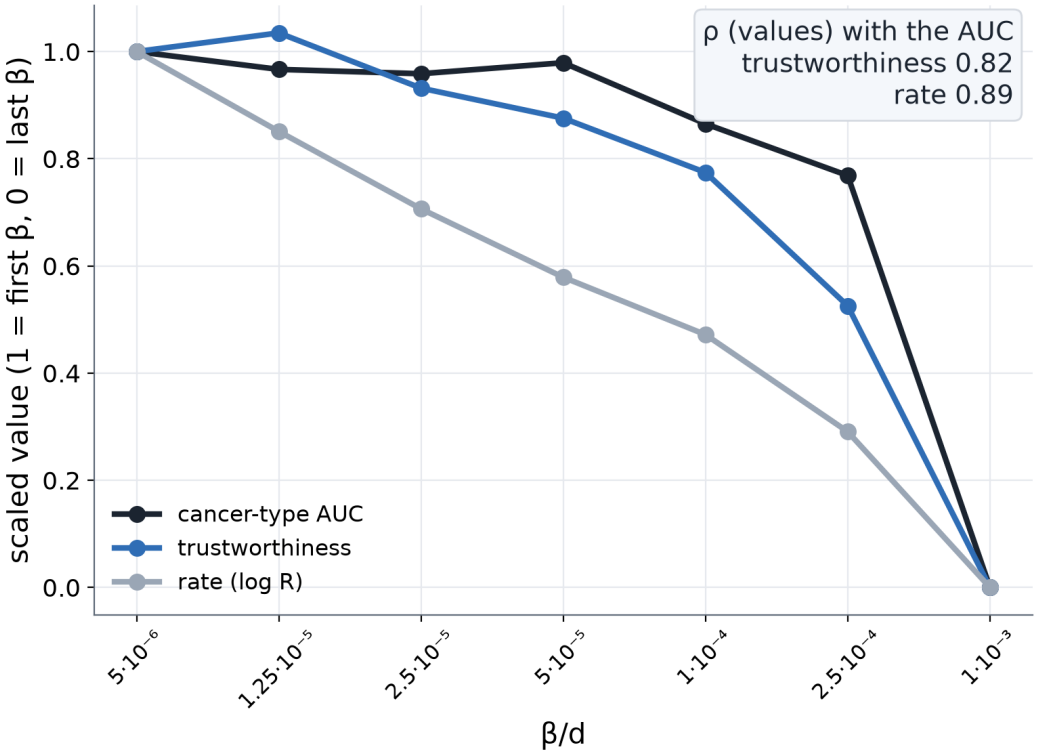
Blue: do they have their big drops at the same β steps? Trustworthiness and LCMC (0.75–0.80) more than the rate (0.60) and for sex all values are low (≤ 0.35).

How computed: per trajectory (one d and seed; 24 = d 2–16 × 4 seeds), states with β > 0 before collapse; Spearman ρ between metric and AUC of the values (dark) and of the drops per log-β step (blue) · median over the 24 trajectories · grey: the rate (log R) · AUC on z (one draw), sex only within COAD, LUAD, LUSC, READ.

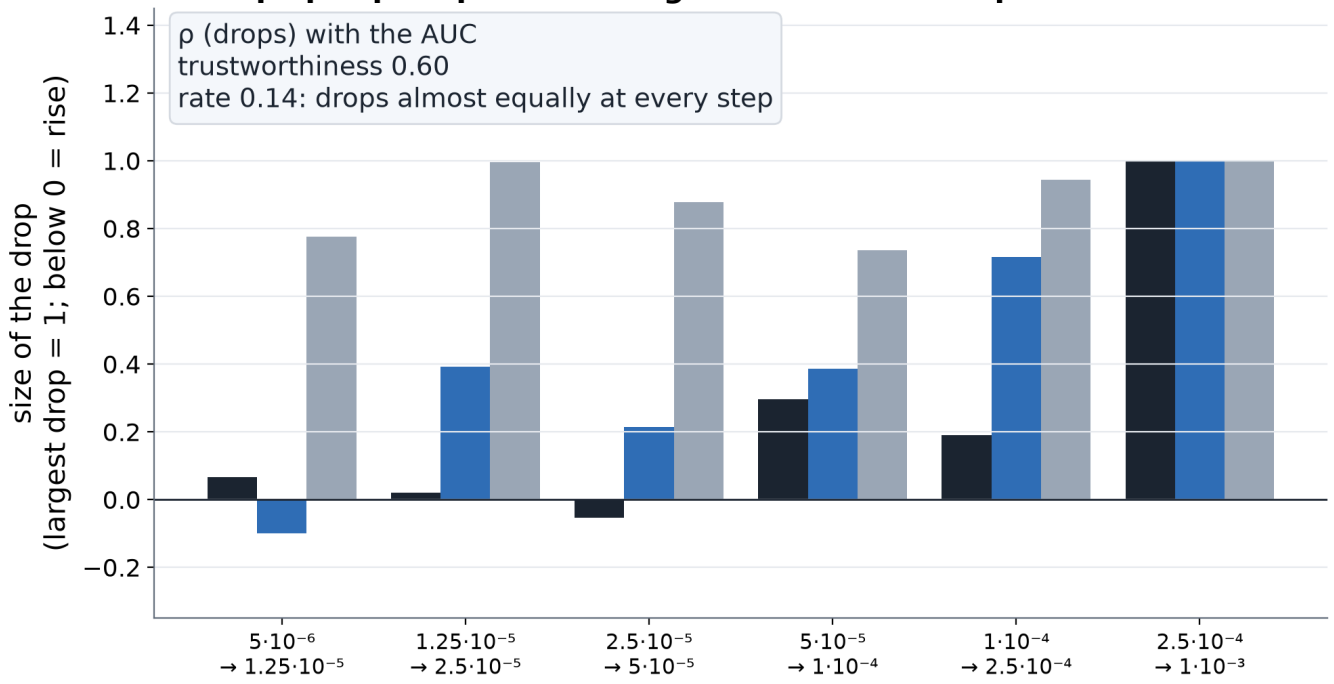
One seed across β · d = 8, seed 17

Values: ρ = 0.82 for trustworthiness, 0.89 for the rate -> only the drops tell them apart (0.60 vs 0.14)

1 Values: all three fall (almost)



2 Drops per β step: are the big ones at the same place?



Left: Spearman of the values → anything that keeps falling scores high, the rate too. Right: Spearman of the drops per β step -> high only if the big drops sit at the same steps as the AUC's.

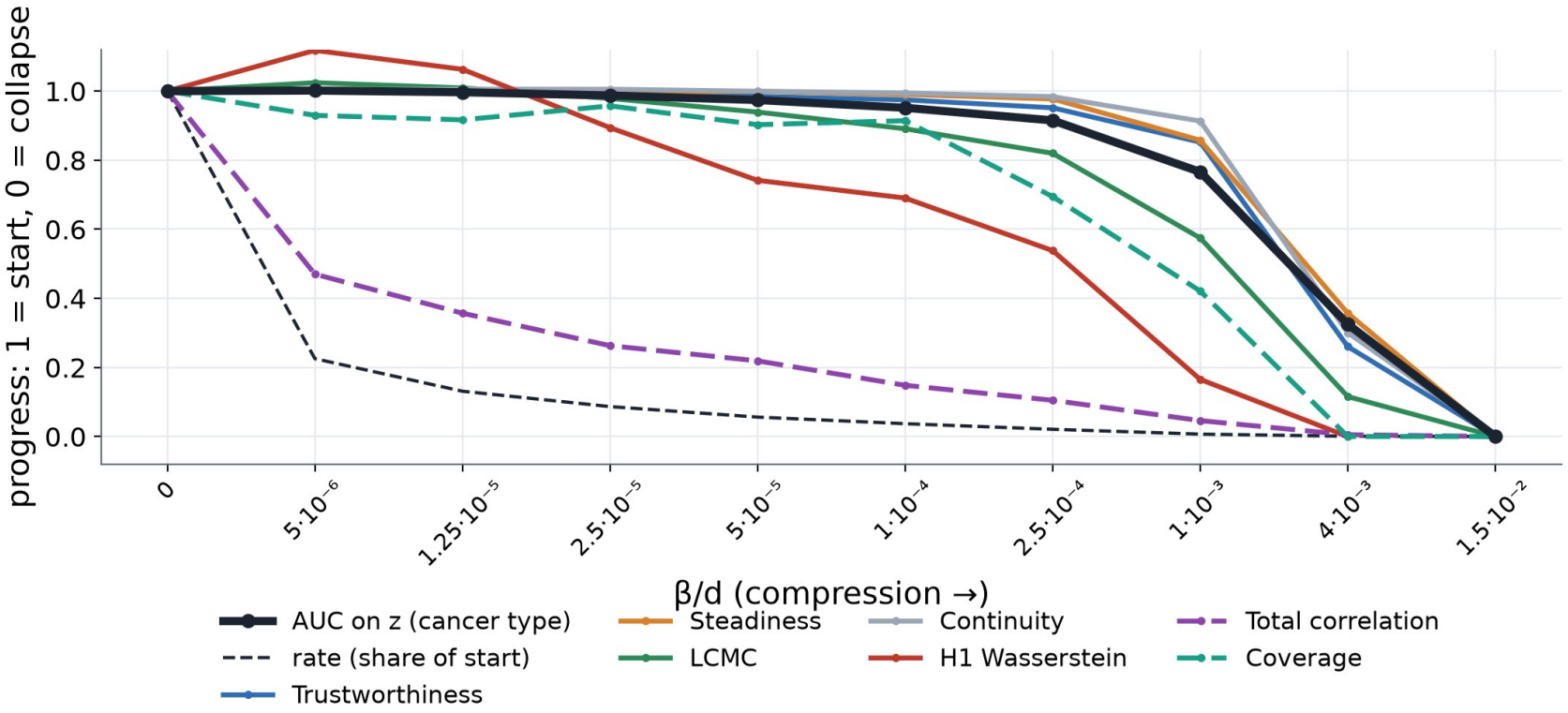
How computed: one trajectory (d = 8, seed 17), its 7 states with β > 0 before collapse · cancer-type AUC on z (one draw, validation) · trustworthiness on μ (test) · rate = log of the mean KL per sample · the bars on "The metrics · both tasks" are the medians of these numbers over 24 trajectories.

Next: In which latents are which informations? Can it helping controlling the beta?

TASK A · CANCER TYPE

Across β · AUC on z (one draw)

Preservation curves and AUC have different transition regions in this sweep



Median over 20 trajectories (d ≥ 4); 1 = start, 0 = strongest compression.

H1 Wasserstein (loops, x vs. μ): −5 % at $\beta/d = 2.5 \cdot 10^{-5}$, a decade before the AUC.

How computed: per trajectory (one d and seed) 1 = β/d = 0, 0 = strongest compression · median over 20 trajectories (d = 4, 8, 12, 16 × 4 seeds) · AUC on z (one draw), validation split.

TASK B · COLLAPSE (RATE)

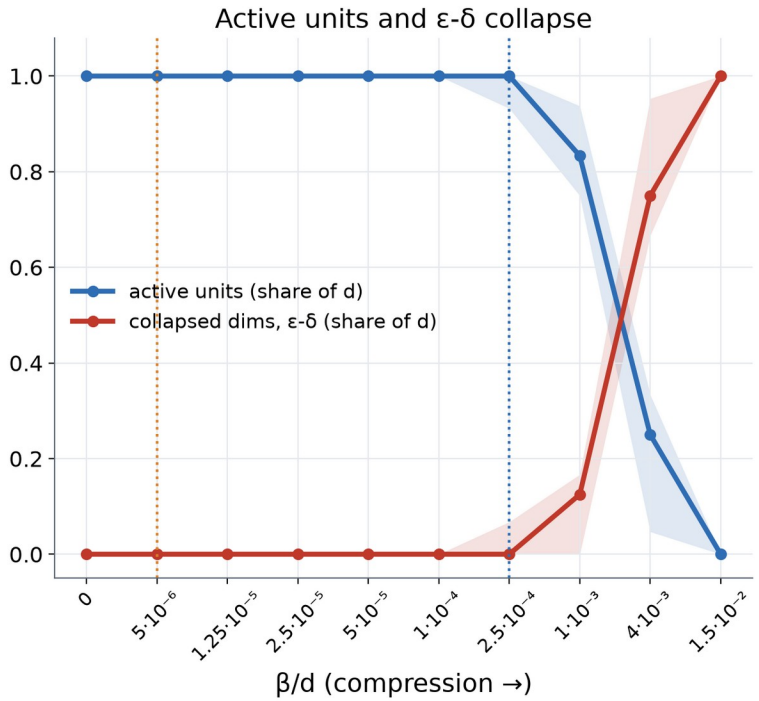
Which metrics explain the rate, and posterior collapse?

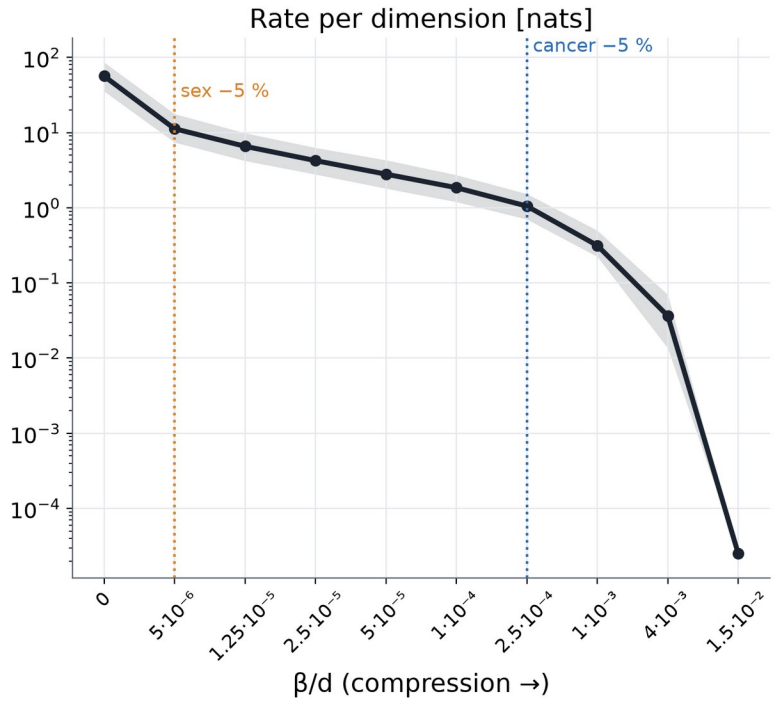
Short answer: the intrinsic dimension of z, graph clustering and LCMC follow the rate almost exactly - both tasks are lost while every latent unit is still active.

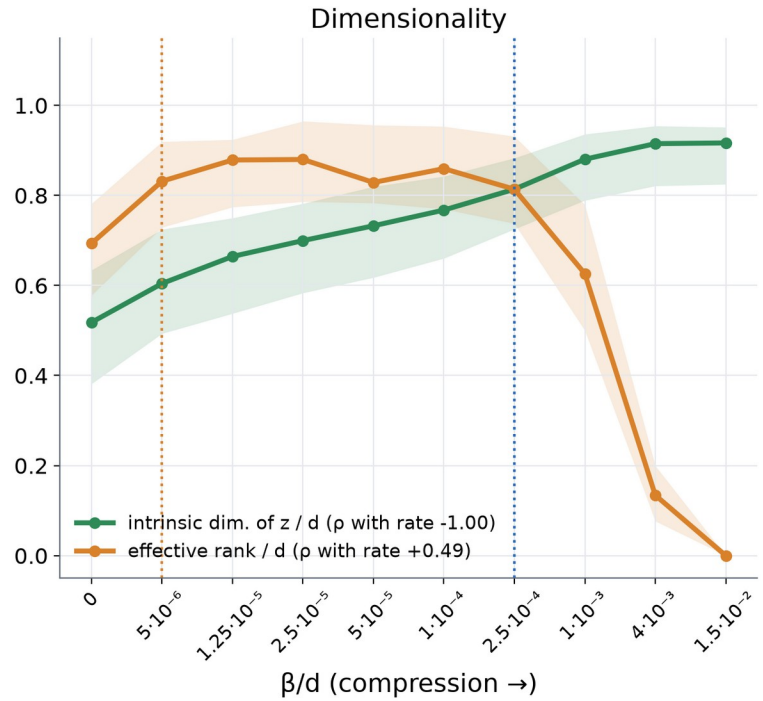
POSTERIOR COLLAPSE

Across β · rate from the encoder

Task performance can decline before latent units become inactive







$\beta/d = 1 \cdot 10^{-3}$

first inactive unit (variance of a latent mean ≤ 0.01)

$2.5 \cdot 10^{-4} \cdot 5 \cdot 10^{-6}$

AUC −5 % (cancer type · sex): all units still active

ρ = -1.00

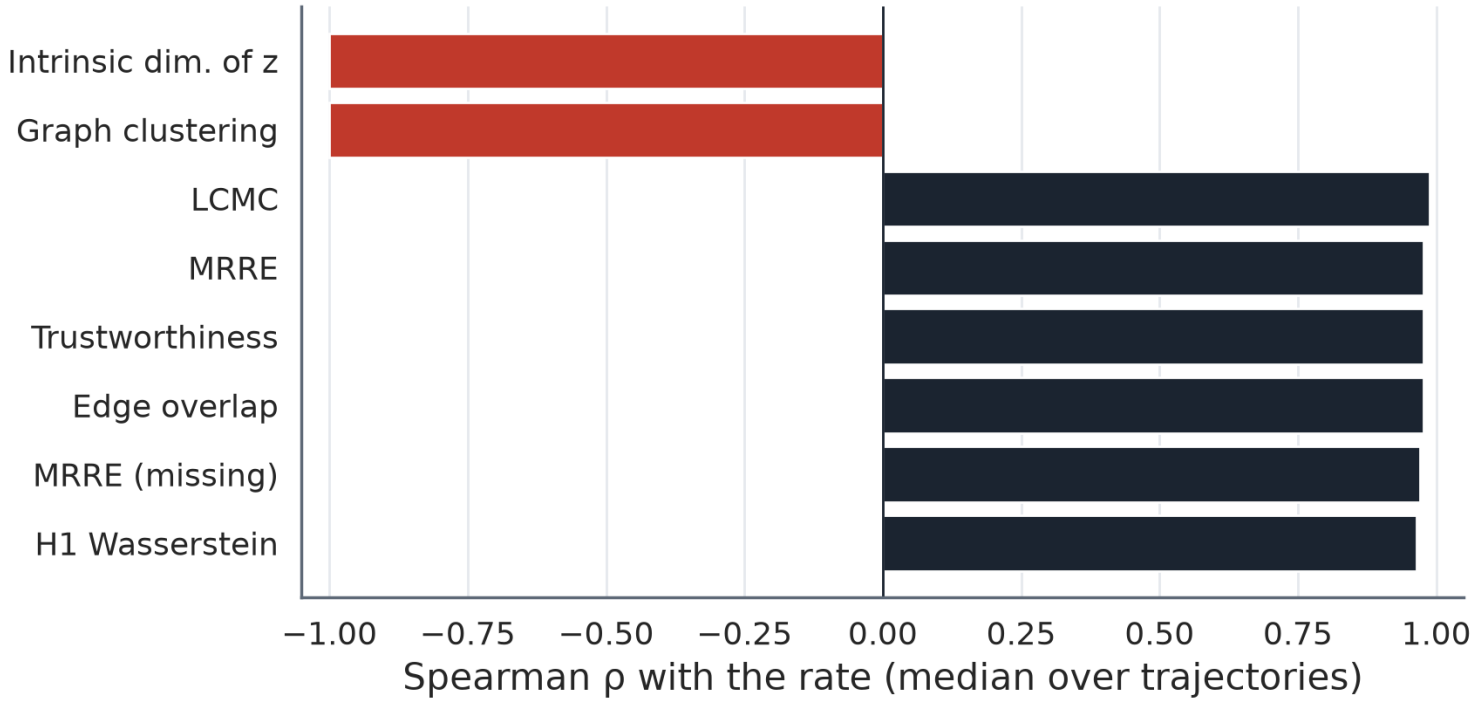
intrinsic dim. of z vs. rate: the best rate tracker

How computed: median over 20 trajectories (d = 4, 8, 12, 16 × 4 seeds), band = IQR · active unit: variance of the latent mean over samples > 0.01 · rate: stored mean KL per sample.

TASK B · COLLAPSE

Across β · rate from the encoder

z dimension, graph clustering and LCMC follow the rate almost exactly for collapse



0.60

R^2 of the rate from active units and ε-δ collapse (step functions)

0.93

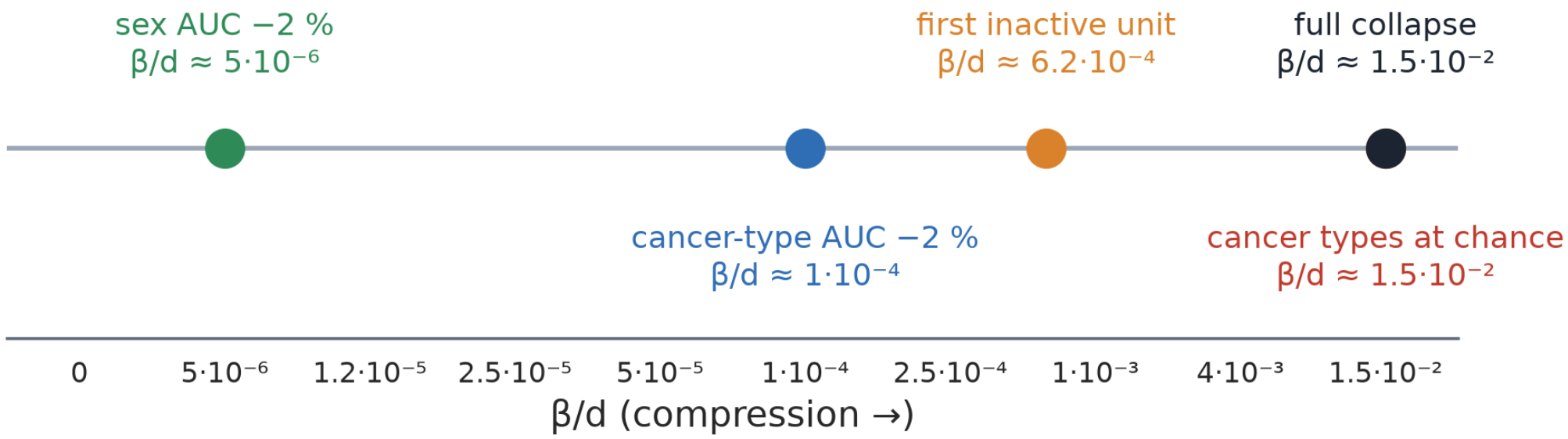
R^2 of the rate from four metrics (LCMC, effective rank, z dimension, betweenness)

How computed: Spearman ρ with log rate within each trajectory (one d and seed, across β), median over trajectories (all d × 4 seeds).

COLLAPSE VS. TASK LOSS

Across β · AUC on z (one draw)

The task is lost long before the model collapses, and sex much earlier



β/d ≈ 5·10⁻⁶

sex AUC loses 2 %

β/d ≈ 1·10⁻⁴

cancer-type AUC loses 2 %

β/d ≈ 6.2·10⁻⁴

first inactive latent unit

β/d ≈ 1.5·10⁻²

full collapse

How computed: per trajectory the first β/d at which each event occurs (AUC loss relative to β/d = 0 of the same d and seed) · median over 20 trajectories (d = 4, 8, 12, 16 × 4 seeds) · AUC on z (one draw).

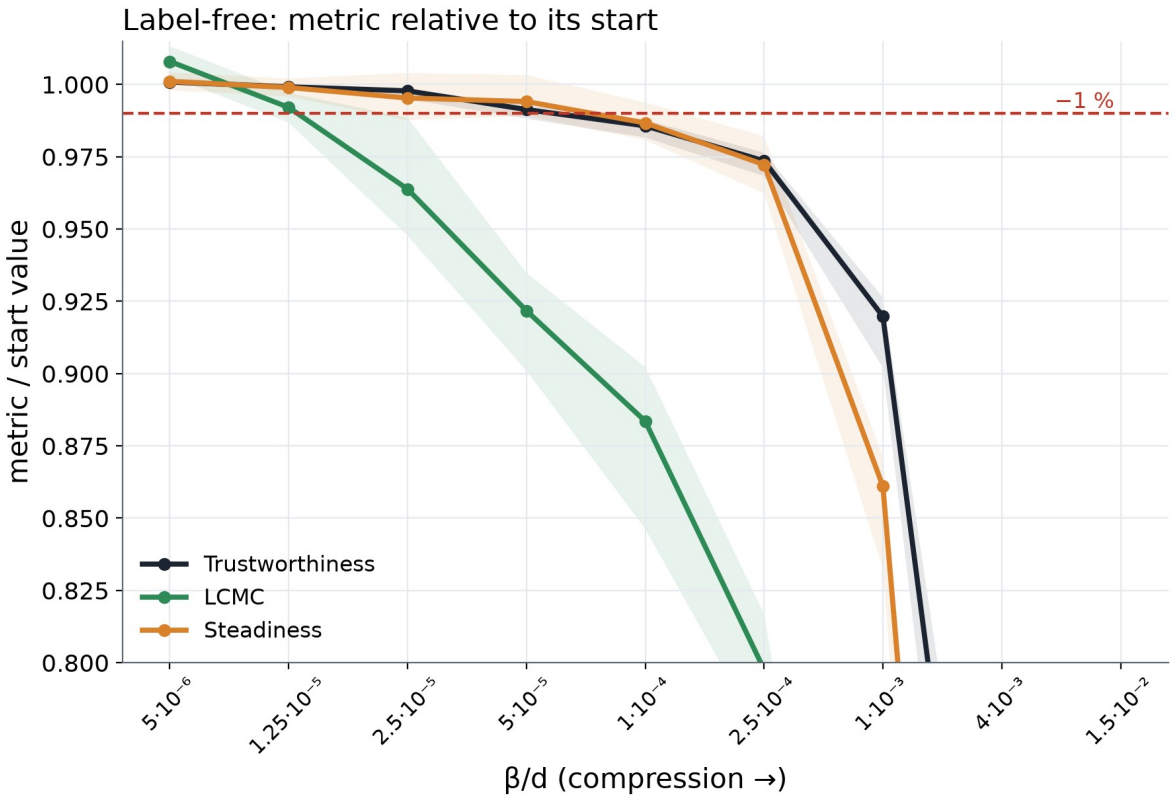
PART 3 · STOPPING

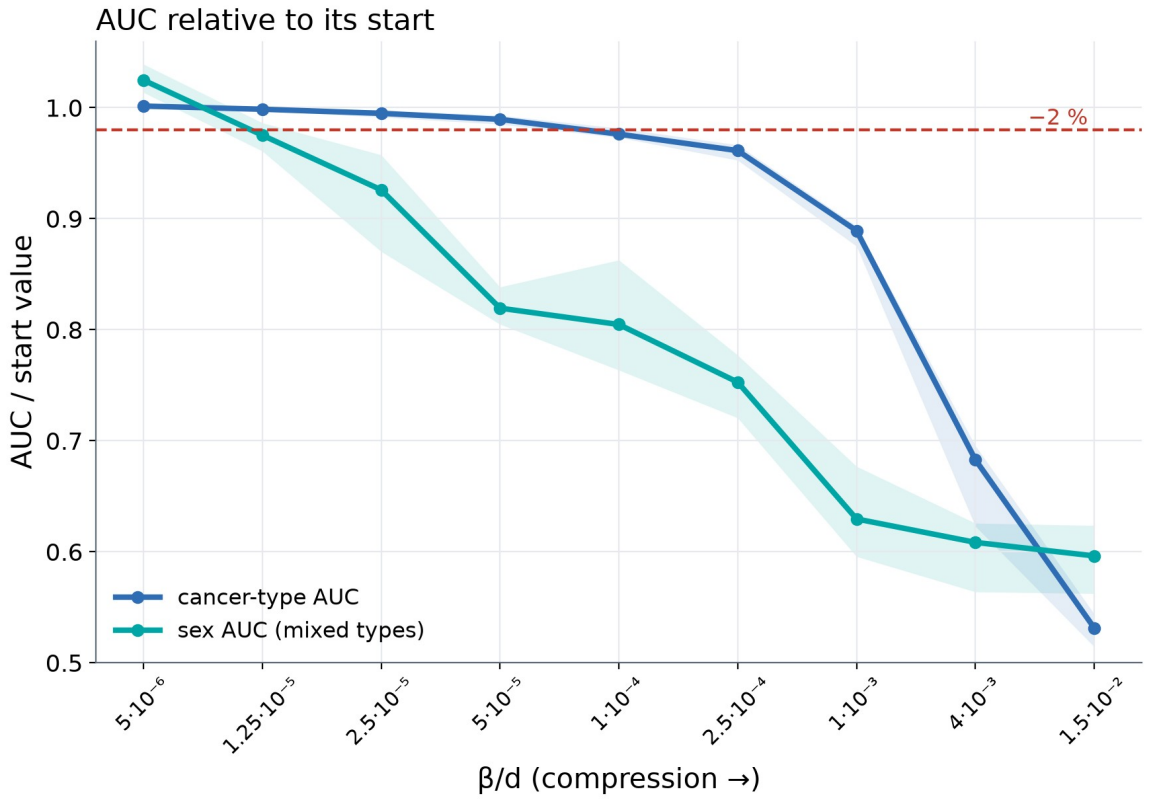
Can a metric/evaluation support a calibrated stopping rule?

A −1 % metric rule keeps the cancer-type AUC in this sweep

Across β · median d ≥ 4

Example thresholds

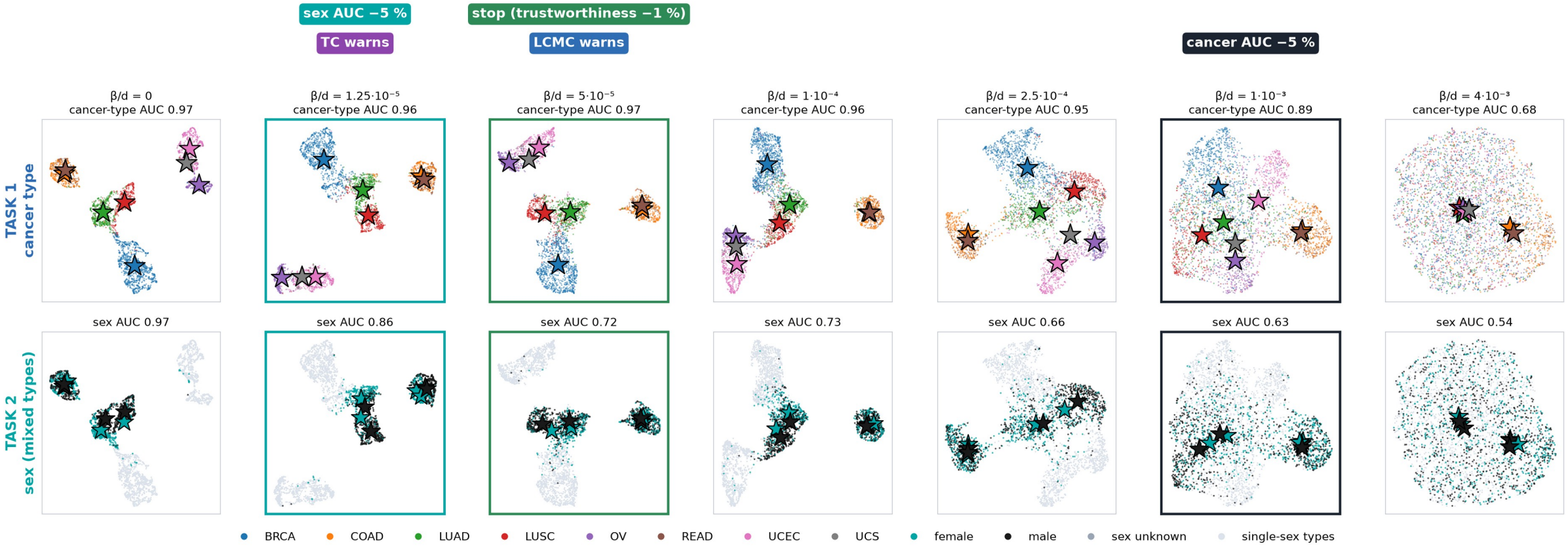




How computed: each β a separately trained model · start = mean of the first two compressed models (β > 0) · median over 20 trajectories (d = 4, 8, 12, 16 × 4 seeds), band = IQR · AUC on z (one draw). The −1 % and −2 % lines are example thresholds

One seed across β · d = 8, seed 17

One trajectory: the warnings fire while the cancer-type clusters still look intact

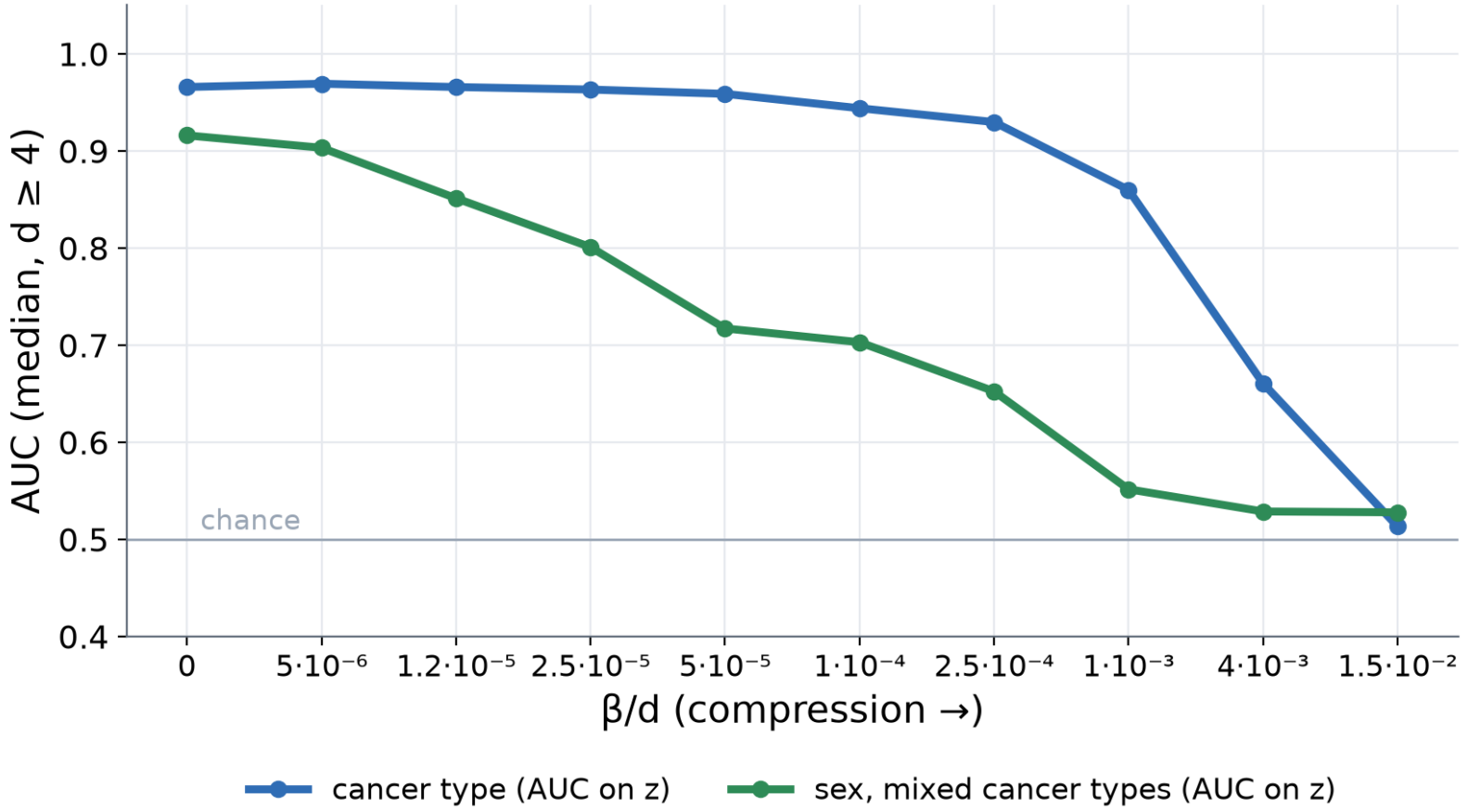


Total correlation warns at $\beta/d = 5 \cdot 10^{-6}$, LCMC at $5 \cdot 10^{-5}$ and trustworthiness −= 1 % stops at $5 \cdot 10^{-5}$; the cancer-type AUC loses 5 % only at $1 \cdot 10^{-3}$, sex already at $5 \cdot 10^{-6}$.

Bottom row grey: single-sex cancer types (BRCA 1072 women / 12 men; OV, UCEC, UCS no men) and mid grey: sex unknown. Both are left out of the sex AUC.

Across β · AUC on z (one draw)

Sex AUC declines earlier than cancer-type AUC in the latent representation

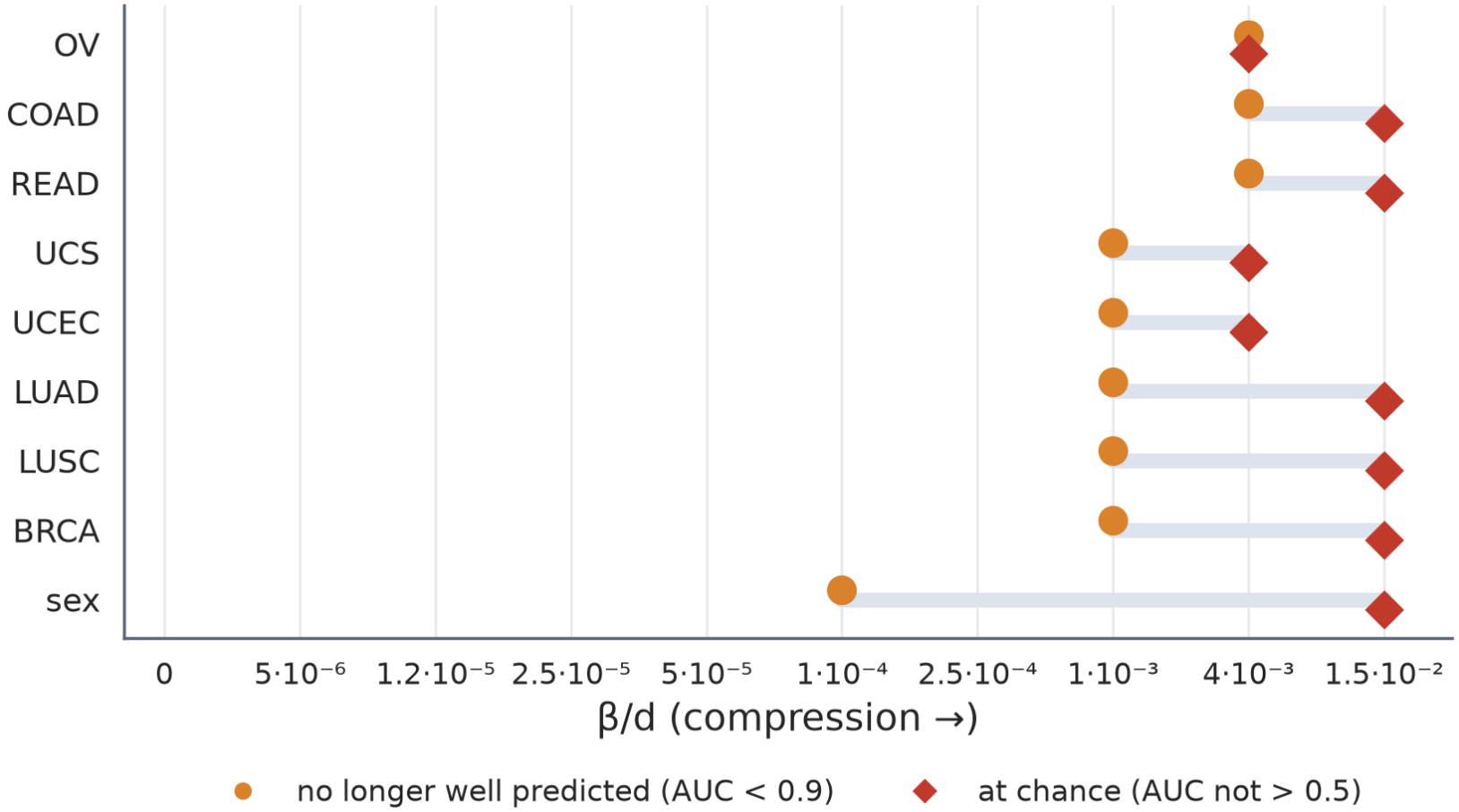


How computed: median per β over 20 trajectories (d = 4, 6, 8, 12, 16; 4 seeds each) · AUC on z (one draw): cancer type validation, sex valid + test within COAD, LUAD, LUSC, READ.

CLASSES

Across β · AUC on z (one draw)

Class by class: each class first becomes hard to recognise, and only at full collapse it is at or below chance-level AUC



Each row: one class, recognised against all others (one-vs-rest AUC).

Orange dot: from this β/d on the class is no longer well recognised (AUC < 0.9).

Red diamond: from here on it is at or below chance-level AUC (AUC not above 0.5).

Sex is the first to become hard ($5{\cdot}10^{-5}$ … 10^{-4}) and most cancer types follow at 10^{-3}

Near-chance AUC occurs mainly at the strongest compression

How computed: one-vs-rest AUC per class (logistic regression on one z draw, fit on train, valid + test pooled) · per trajectory the first β/d of each event · median over 20 trajectories (d = 4, 8, 12, 16 × 4 seeds).

SUMMARY

AUC on z (one draw)

Exploratory candidates by purpose; no metric has a validated universal threshold

Purpose	Cancer type	Sex (mixed types)
Early warning (stop)	H1 Wasserstein, LCMC, intrinsic dim. of z, total correlation	none; at best total correlation, intrinsic dim. of z
Model choice at the same β	Continuity, trustworthiness, LCMC, stress	LCMC, graph transitivity, partition NMI, graph clustering
No demonstrated utility here	Rate proxies (intrinsic dim. of z, total correlation)	Trustworthiness, steadiness, continuity

CONCLUSION

Rate measures compression; preservation may help select useful representations

Rate for collapse

Preservation for task

At the same β, continuity, trustworthiness and LCMC show exploratory associations with cancer-type quality (4 seeds per cell).

Weak signals

Sex is sacrificed at the first β steps and structure metrics warn too late. Finding and Analysing important genes that express the different sex could be interesting

OUTLOOK

Limitations and next steps

Limitations

- One dataset (TCGA), four seeds per dimension.
- β/d grid is coarse between 0 and 10^{-4} (sex) and above 10^{-3} (classes).
- Observational but no causal proof.

Next steps

- furhter β grid between 0 and 10^{-4}.
- Validation on geometric datasets with known structure (circle, swiss roll).
- Optuna for the optimal β/d under the preservation constraint.
- Way more epochs, lr, dataset splits etc. for vae reaction to capacity change
- Controlling beta and also look what the model does after lowering beta again.
- Dimension wise beta adjsutment

NEXT EXPERIMENTS

Next: validate metric–AUC associations at fixed hyperparameters across seeds

	Current	Next (fixed β)
Latent size · seeds	d = 2–16 · 4 seeds	d = 4, 8, 16 (AUTOENCODIX default 16) · 12 seeds
β/d	10 levels, 0 … $1.5 \cdot 10^{-2}$	0, $5 \cdot 10^{-6}$, $2.5 \cdot 10^{-5}$ (sex) · 10^{-4}, $2.5 \cdot 10^{-4}$, $5 \cdot 10^{-4}$, 10^{-3} (cancer type)
Training · data	90 epochs but could still improve· one daat split	250 epochs (pilot to 400 first) · two data splits (seeds 2 and 7)
Downstream	AUC on one z draw	10 z draws: mean AUC from 10 probe fits
Metrics	56, exploratory	a short list fixed in advance
Geometric data	–	circle, swiss roll, sphere; latent d 1–3; truth: Betti numbers, dimension, neighbourhoods

Goal: minimise rate while retaining specified tasks

Next 2: validate a preservation-constrained rate controller

